



CSE-4508

Relational Database Management System

Lab 8

Lab Group: **1A**

Thursday, 5th February, 2026

1 E-Commerce Analytics with ROLLUP and CUBE

Task 1.1

Write a query using ROLLUP to generate a hierarchical summary of total sales revenue by region, country, and city. Include subtotals at each level.

Task 1.2

Use CUBE to analyze sales across category, subcategory, and region simultaneously. Calculate total revenue and average discount for all possible combinations.

2 Date Operations

Task 2.1

Write an SQL query to display the transaction date in the following format:

Day, DD Month YYYY
(e.g., Monday, 15 January 2024)

The output must include the following columns:

- `transaction_id`
- `formatted_transaction_date`

Task 2.2

Using the `TO_CHAR` function, write a query to group transactions by month and year. For each month, calculate:

- Total number of transactions
- Total sales revenue after discount

The output must contain:

- `month_year` (e.g., Jan-2024)
- `total_transactions`
- `total_revenue`

Task 2.3

Assume that all products under the *Electronics* category come with a warranty period of 12 months.

Write a query to display the following for each Electronics transaction:

- `transaction_id`
- `transaction_date`
- `warranty_expiry_date`

The warranty expiry date should be calculated using the `ADD_MONTHS` function.

Task 2.4

Assuming the current date is 2024-03-01, write a query to calculate the number of days since each transaction.

The output should include:

- `transaction_id`
- `transaction_date`
- `days_since_transaction`

Task 2.5

Write a query to find all transactions that occurred more than 30 days before 2024-03-01.

The output should display:

- `transaction_id`
- `transaction_date`

3 Regular Expression Pattern Matching

Task 3.1

Write a query using regular expressions to identify all records with invalid email addresses. A valid email should match the pattern: `username@domain.extension`

Task 3.2

Use regex to extract titles (Dr., Mr., Mrs., Ms., Prof., etc.) from customer names. Create a separate column showing the title and the name without the title.